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Graduate Challenge 2026

Business Challenge Scenario

Brief

Wallaby Mining (WM) is an ASX-listed Australian mining & exploration company.

The company was formed in 2016 and subsequently acquired an 60% shareholding of the Waratah Resources Joint Venture (JV). At the time, weakened coal price created an opportunity for the company to acquire a share in the struggling JV, with the purchase conditional on assuming the attributable closure liability expected from 2032 onwards. Until this stage, the company would be entitled to 60% of the revenue derived from the utilisation of the pre-existing infrastructure. As a part of the transaction, WM has acquired one (1) operational underground Coking Coal mine, one (1) historic open cut that produced a ~70/30 thermal to coking coal split that has since ceased mining due to reaching lease limits and one (1) underground project.

These are:

- Springbok Coal Complex (currently operational)
- Brumby Open Cut (historic)
- Brave Blossom Project (underground)

The Springbok Coal Complex has a long and storied history, with first coal mined at the Brumby open cut in 1992, followed by underground operations from 2000. The open-cut reserves within the ML are now depleted however the resource extends beyond the ML into the neighbouring mining lease at a similar strip ratio and quality to what was historically mined. The underground operations are progressing deeper, becoming gassier and more geotechnically challenging. Springbok underground is now 5 years from closure and there is a heightened focus on minimising the end-of-life liability.

The Wallaby Mining board have requested through their annual Life of Mine process to review the businesses valuation, with the primary objectives being the following:

1. Strategic assessment of the Wallaby Mining portfolio

The board is eager to provide an update to shareholders (of both the company and the Waratah Resources Joint Venture (JV) on the long-term strategy of unlocking the full value of the Springbok Coal Complex and its associated project. As the management team of WM, you are to utilise the latest geological information and concept study provided to conduct this

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comprehensive review. Following this, you are to present your team's analysis to the board and make a recommendation for the best strategy to maximise shareholder returns beyond the current mine life including any further projects that could be progressed to a concept study.

2. Review and recommend improvements to Springbok Coal Complex's closure estimate and post mining liabilities

The Board has requested that the site team evaluate the current operations closure estimate and post-mining liability and present their recommendations to minimise the post-mining liability, including underground decommissioned mine emissions.

Current Operations

The Springbok Coal Complex is located approximately 80km North-West of Emerald within Queensland's Bowen Basin. The Barada People hold registered native title over the land on which the complex operates. Springbok mine is a small underground operation that utilises 1 x longwall and 3 x development miners to historically produce 3.5mtpa of premium hard coking coal. The underground operations have successfully mined and marketed this coking coal for decades and the product is proven and preferred by WM's Japanese customers (including members of the Waratah Resources Joint Venture (JV)).

Challenges have been encountered in recent years with worsening geotechnical conditions and gas drainage impacting the development schedule. A discontinuity in the longwall schedule has resulted from this with a further park up in 2027, followed by minimal longwall float for the remaining life of mine.

Springbok coal has historically had an agreement with a local energy producer to send the gas offsite for power generation with this contract set to expire at last coal if clauses are not triggered prior.

Projects

Wallaby Mining has a highly prospective project at concept level with drilling suggesting there is further future projects yet to be assessed. Initial drilling programs have been completed at the Brave Blossom project site, but there is a requirement for a further resource definition and pre-feasibility studies. Considerations of the lead time required for studies, approvals, infrastructure, and equipment, suggests if this project is to progress, the forward work plan will need to be fully funded and commencing in 2027 to ensure continuity of cashflow beyond 2030 on the cessation of Springbok Underground. All forward work programs must have an accompanying business case (financial model) to support the capital spend.

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Historic mining suggests there are further targets that could be considered and progressed once a concept study had been completed.

Brave Blossom Mine UG – Concept

There are old underground workings close to Springbok that were previously worked in the early 2000's but eventually sealed due to persistent roof falls. There remains a significant resource to the East of the workings that while large in size, were previously considered too geologically complex and unsuitable for longwall mining. Following a minor drilling program showing more favourable structural conditions, a new mine plan was proposed where the workings will be re-entered to establish a longwall operation capable of producing ~6mtpa of saleable hard coking coal until 2052. Although the drilling program provided favourable geotechnical conditions the Coal Quality has deteriorated requiring further marketing considerations.

Currently ~75% of the current mine plan is contained within the mining lease, with the reminder ~25% currently sitting under an MDL. It is expected that a range of federal and local government approvals will be required to support a significant amendment to the current approvals. While State approval exists to mine the area within the current ML, there remains uncertainty regarding the ability to secure the necessary federal approvals.

Other Considerations

CHPP

The CHPP (design capacity feed 8Mt/year) and TLO are located close to current underground operations. The coal from Springbok is currently conveyed from the underground to the CHPP, where it is processed into a Premium Hard Coking product (~4mtpa saleable @ ~67% yield). Rejects are hauled to a nearby pit void, whilst tailings are pumped into an above ground tailings storage facility adjacent to the CHPP.

There current CHPP is expected to reach its end of life in 2032 if the current throughput rate is maintained. The plant will require major overhauls and replacement components if there is any consideration to extend the life of the asset including new conveyor, major overhauls of the raw coal bins and significant structural repairs to the plant.

Logistics & Marketing

All existing infrastructure for port and rail will continue to be utilised as per the arrangement at present. The current rail and port contracts arranged in 2025 were for 4mtpa through until 2028. The recent underperformance and associated inability to fulfill this quota has resulted in a carried take or pay (TOP) liability, increasing the DSE cost by \$10/saleable tonne. The WM's logistics team are currently in negotiation for the 2028-2031 period, where initial

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indications are that for the same volume, costs will remain in line with the current arrangements. However, the introduction of 2 x large new competing mines onto the line will remove any residual capacity of the network from 2028. Following the recent weakness in the coal market, the WM marketing team have also provided their revised forward pricing curves within the valuation model provided. The logistics team are required to provide the port and rail providers with a renewal request detailing tonnage requirements 1 year prior to expiry.

Carbon

WM is actively assessing the potential effects of carbon pricing mechanisms and evolving environmental regulations, particularly with the forthcoming carbon tax reforms, which are expected to impact coal exports starting in 2027. The WM gas assignment model encompasses both the current operations and proposed project site. Scope 1 emissions are primarily driven by fossil fuel combustion, post-mining activities, coal mine waste gas, and emissions from decommissioned mines, with estimated emissions detailed in the valuation model.

Under the Safeguard Mechanism, facilities that emit over 100,000 tonnes of carbon dioxide equivalent annually are subject to regulations. Any emissions exceeding the baseline level (as outlined in the valuation model) will require abatement or will need to be offset through the purchase of carbon credits.

Decommission mines as defined by the National Greenhouse and Energy Reporting Determination (NGER Determination) are mines in which coal production and active mine ventilation have ceased to occur and are not expected to occur in the future. These emission will continue until; 20 years after the cessation of the above activities has occurred. This can represent a material liability, if no alternative carbon-emission-reduction strategies are investigated and implemented.

Mine Rehabilitation & Closure

Springbok is a large complex, with numerous pit voids left on completion of mining that are currently be utilised for water and tailings storage. As a socially responsible mining company, avoiding the rehabilitation on completion of mining activities is not an option the board is willing to consider. Regardless of the future of the business, the current estimate relating to legacy rehabilitation is A\$900M, which is considerably higher than what was anticipated at the time of purchase. This estimate was developed by a third party using a calculator based on the QLD ERC estimate tool using unit rates and activity-based costing agreed between the site and the third party.

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Progressive rehabilitation of newly disturbed areas is also being completed for ~\$2.5M / annum. Mine closure is currently scheduled to commence following the cessation of operations at the end of Q4 2031. Completion of mine closure and relinquishment of the mine leases back to the government is currently expected to occur by 2050.

Social

As Wallaby Mining has long been the major supporter of the local government and town where the workforce resides. It is anticipated that if activities were to discontinue beyond Springbok, the town and local economy that is nearly entirely dependent on the complex would be highly impacted. There has been significant media attention recently on mining towns, with a competing mining company in the area facing media backlash for closing a mine site and discontinuing support for the town and its residents. Procurement is also struggling to maintain supplies to the operation as local companies are now diversifying their business to be less reliant on mining clients causing increased lead times on parts and supplementary labour.

Neighbouring Mines

The Springbok coal complex shares the northern lease boundary with Bronco mining which has a small opencut operation that is currently operating. Wallaby Mining have approached the Bronco Mining board on several occasions to purchase the struggling opencut however they have been rejected numerous times. Bronco Mining operate another opencut in the area have negotiated a deal with a neighbouring mine for a royalty-based extraction of the resource within their lease.

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Your Task

Firstly, conduct a strategic review of the life extension options within the Springbok Coal Complex (i.e., Brave Blossom Underground & any future prospective projects or extensions). It may also be prudent to consider the option of not extending beyond the current mine life at Springbok. As the Brave Blossom project is at concept stage, the intent of the review is strategic, and recommendations should therefore be made with this in mind.

Secondly, evaluate current operations and provide clear recommendations on post mining liability mitigation measures. It is suggested that the recommendations adhere to the SMART criteria to ensure deliverability.

In presenting your recommendation, you will be expected to outline the key strengths and weakness as well as any opportunities and threats (SWOT) of each option. You will also be expected to include an indicative forward work plan on how to progress your recommendation to address any key risks. Your presentation should clearly outline your recommendation (backed up by supporting arguments) and should be professional, creative, and influential. It will be up to the group to decide who will present (it can be one person or multiple people) and who will field the questions from the board. Each group will have:

- 15 minutes to present (maximum)
- 5 minutes for questions from the board

The presentation should cover off the following:

- Top 3 recommendations, including:
 - Resultant cashflow analysis
 - Implementation risks
- Strategic Direction Recommendations
 - Valuation and SWOT Analysis
- Recommendation on whether to progress with a forward work plan on the project and when this will occur and recommend any future work to be conducted on additional potential projects

Some items that may be relevant in the SWOT / analysis are summarised below.

- Mine plan physicals & coal quality considerations
- Infrastructure requirements
- Equipment suitability, adequacy, and implications for availability & utilisation
- Geological / geotechnical risk
- Approvals / environmental risk
- Carbon liability
- Deliverability / complexity of the plan

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- Financial implications of the options
- Labour supply considerations (owner operator vs contract miner run)
- People impact including any change management that needs to occur for results of potential structural changes. Consider implications of any changes with respect to work process, roster patterns, housing requirements, health, safety, and training.
- Any other considerations that should be raised with the board

A financial model for both the current underground operation and the project is available. As Wallaby Mining's management team, you must consider what key capital requirements are required to support any project and prepare the capital estimate for them.

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CONCEPTUAL MINING STUDY

Brave Blossom UNDERGROUND PROJECT

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Location

The Brave Blossom underground project is located immediately North-East of the current operating mine within the Springbok Coal complex.



Figure 1 Brave Blossom Project Location

Geology & Geotech

The Mentelle Formation is a sub-unit of the Blackwater Group, and it stratigraphically overlies the barren Ingelara Formation and underlies the Fairhill Formation. Coal seams in the Mentelle Formation are common, thick and well formed; however, only two seams are economically extracted, being Mentelle 1 & 5 seams using both underground and open cut techniques.

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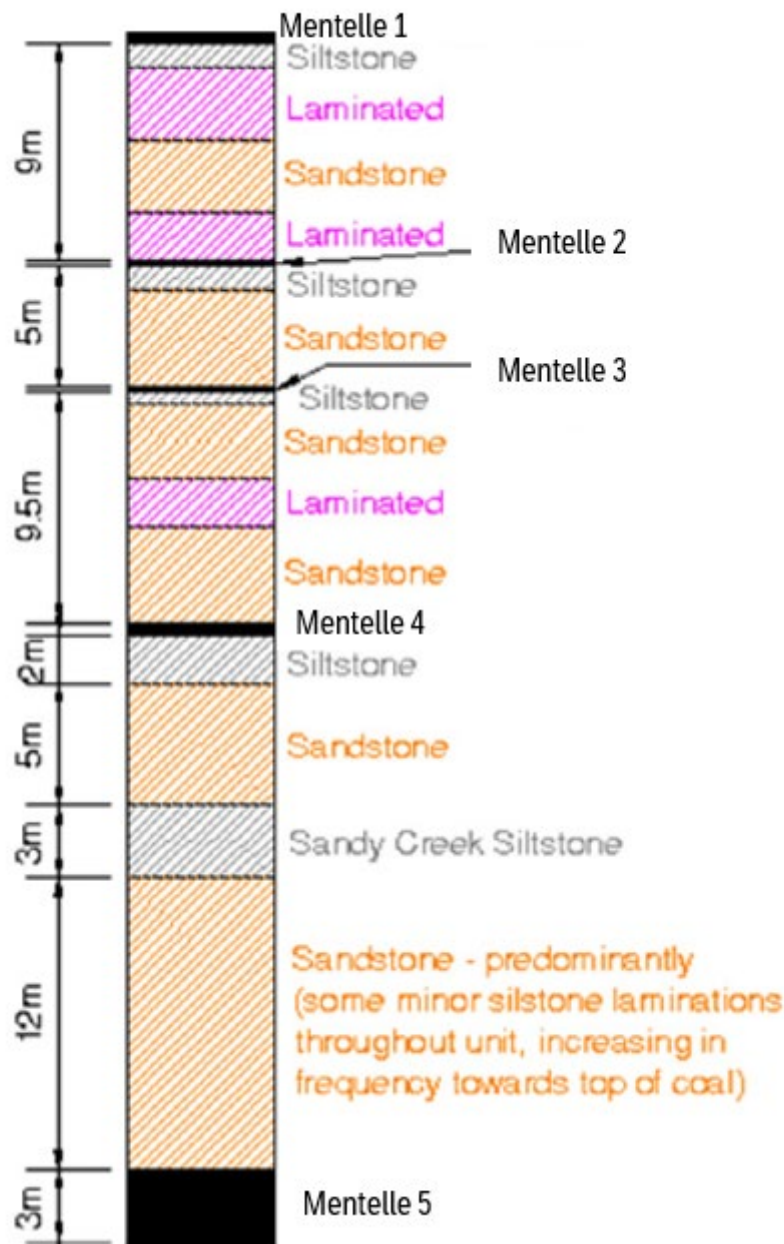


Figure 2 Stratigraphic Section of Area

From the limited information available, the geological conditions anticipated at Brave Blossom are expected to be similar in nature to Springbok Coal. The Springbok underground and open cut mining complex has intersected a range of geological features over the years. In the underground workings both normal and reverse faults have been both successfully and unsuccessfully mined through with the longwall. The orientation of joints and cleats has also required evaluation to optimise the longwall layouts. Other geological features such as monoclines, sedimentary and igneous dykes have also been encountered, with varying impacts to mining.

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A major monocline is present over the mine plan, with localised seam grades of up to 1 in

There is also a similar dome like structure present over the eastern portion of the mine plan. Faulting can be grouped into two dominant fault sets, one trending NNW and the other NE.

Historic Bolt up plans prior to the old working being ceased show high density bolting and the use of mega bolts within the patterns.

Land & Permitting

The land within the project boundary is majority owned by Wallaby Mining. The last 4 panels within the proposed Brave Blossom project area are below privately owned land. Subsidence considerations, landholder consultations and or negotiations are required for these panels to go ahead.

75% of the current mine plan is contained within the current mining lease, while State approval exists to mine the area within the current ML, there remains uncertainty regarding the ability to secure the necessary federal approvals. The reminder ~25% currently sitting under an MDL own by the Waratah Resources Joint Venture (JV).

Working section & Coal Quality

The Mentelle 5 seam is the primary target for the Springbok Coal Complex. The working section thins as the mine plan progresses East, potentially introducing more dilution into the feed.

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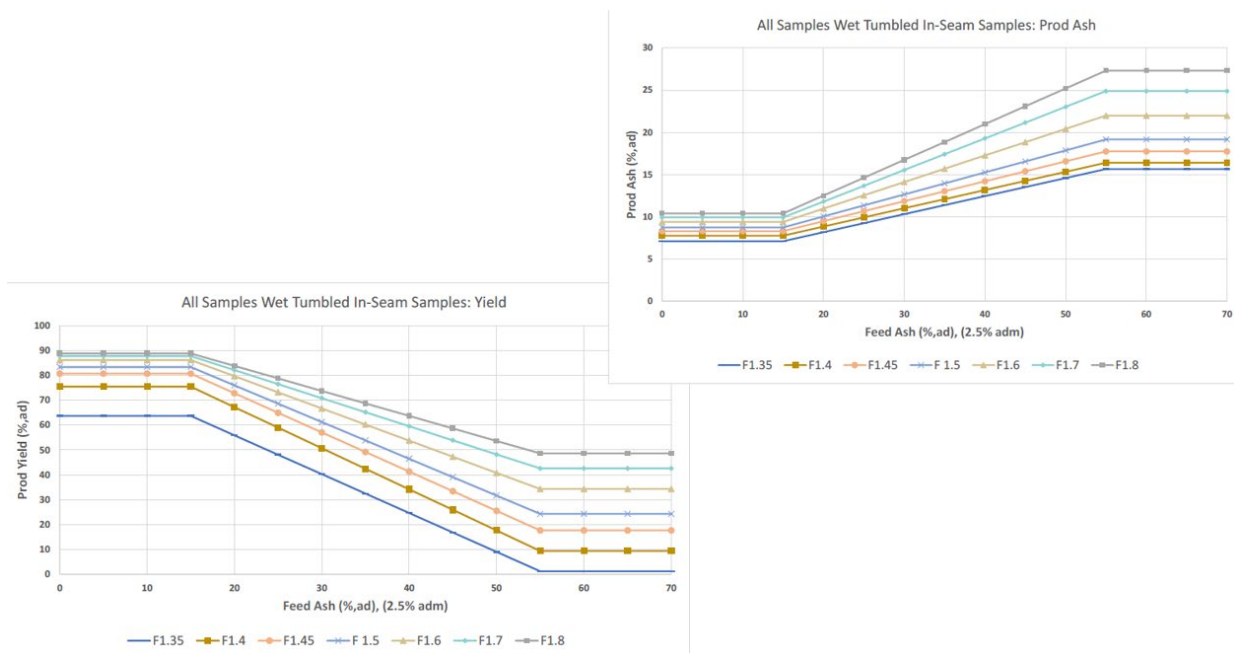


Figure 3 Ash-Yield

Ventilation & Gas

Pre- and post-drainage alone will not be adequate to remove the gas from the workings and a new ventilation network would be required prior to operations. The ventilation network will need to be designed with adequate capacity to dilute residual gas. New intake and exhaust shafts and fans should be located away from the open cut dumps. Leaving a coal roof in the goaf would pose the risk of spontaneous combustion although seams at Springbok have shown relatively low propensity to spontaneously combust when compared to alternative UG operations in the area.

Gas drainage will be required for outburst mitigation and to support high productivity longwall mining.

- Gas content in the Mentelle seam increases with depth, averaging between 12-15m³/t at 300m depth. The gas content of the overlying seams is not known at this stage.
- The methane ratio ((CH₄/CH₄+CO₂)) for all Brave Blossom seams averages 0.98.

The methane content in the Brave Blossom project fall within the stipulated specifications of the contract with the offsite energy company with clauses in the contract to extend this relationship if a project was to be considered.

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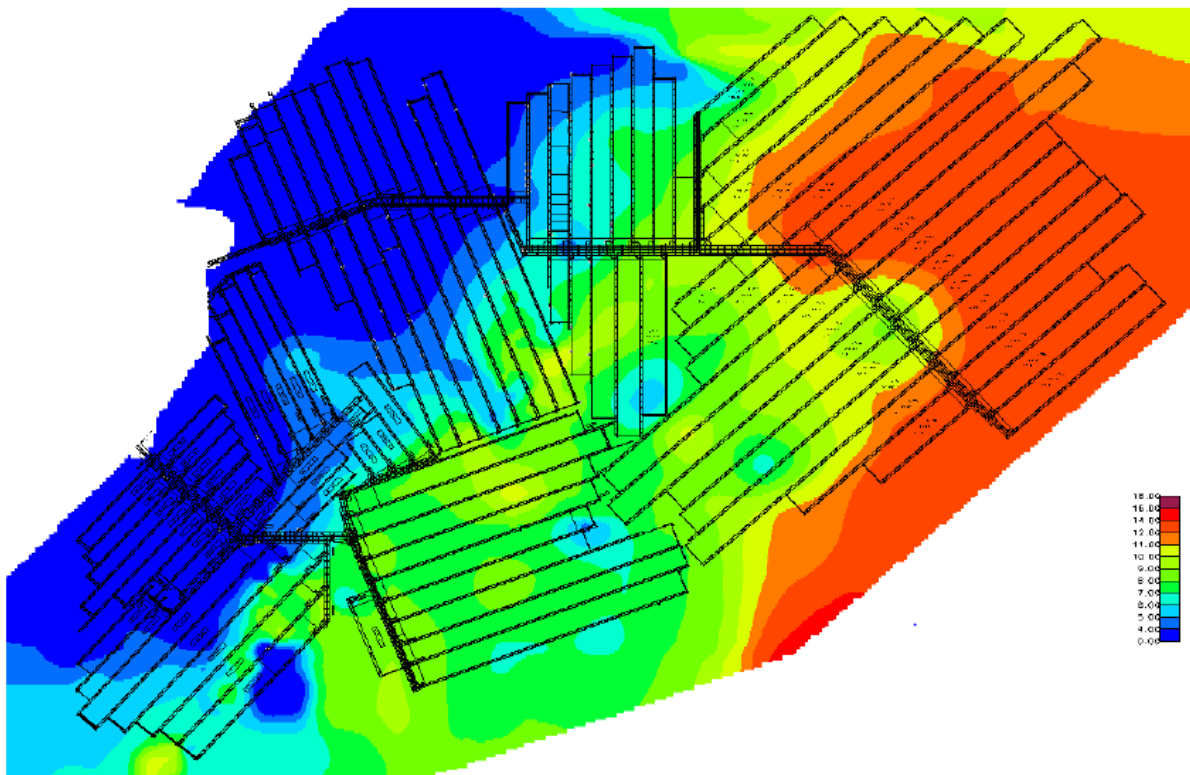


Figure 4 Gas Content

Mine Design

The Brave Blossom UG seam resources have been extracted by longwall operations in the past prior to being sealed following a major roof fall. It is not known at this stage whether water has filled the workings, but the portal seals remain in good shape.

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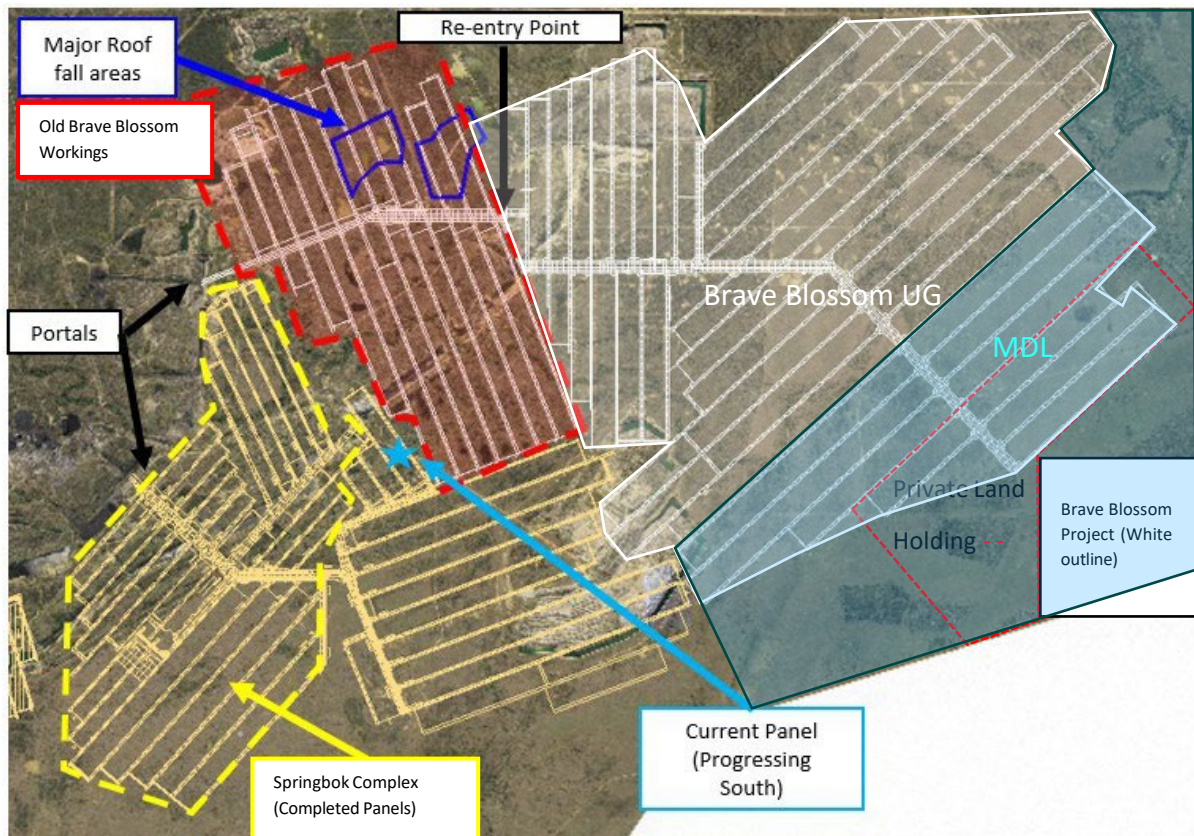


Figure 5 Mine Plan

A re-worked mine plan in the Eastern portion of the resource avoids major structure through orientation and step-arounds and may be able to facilitate high productivity longwall mining. The longwall width has been designed at 300m, supporting an annual production of 8mtpa ROM (walk on – walk off) over the ~20-year mine life.

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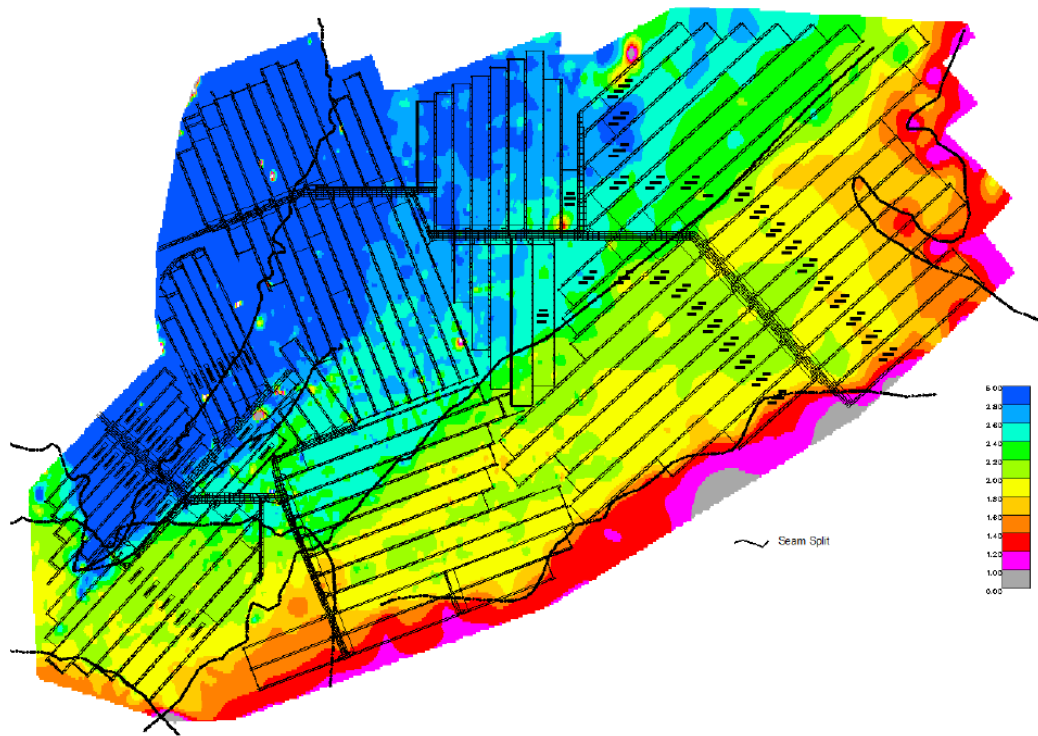


Figure 6 Seam Thickness

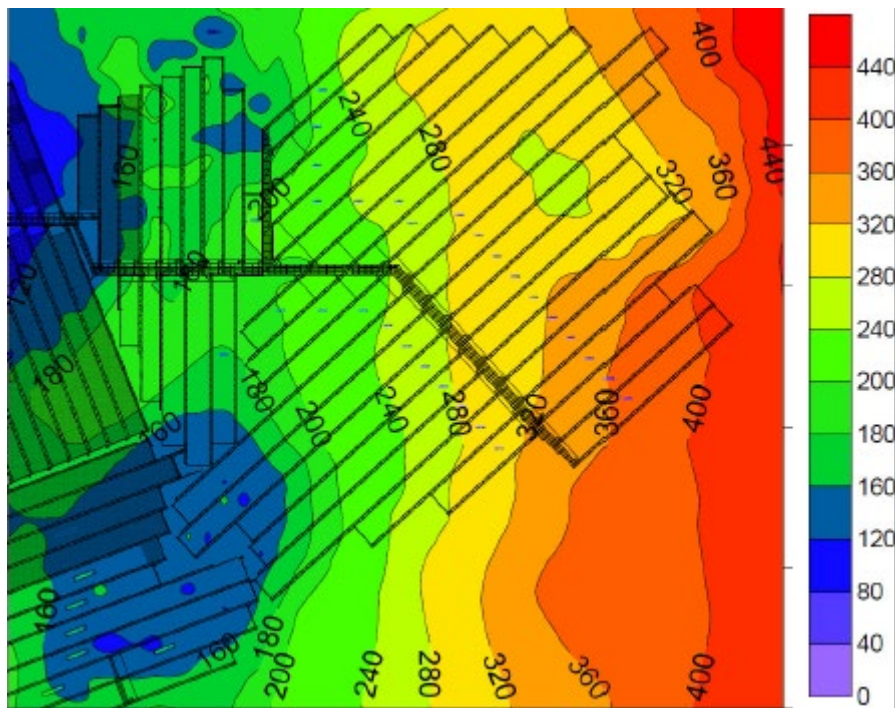


Figure 7 Depth of Cover

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Equipment & Infrastructure Requirements

There is not anticipated to be any residual capacity in most of the equipment from Springbok on the completion of the current LOM. As a result, there is a long list of equipment required to be purchased prior to mining operations. There is limited capacity for pre-existing infrastructure at Springbok (beyond the CHPP) to service Brave Blossom. It is recommended that the management of Wallaby Mining make some high-level assessments of capital requirements (including cost to re-enter workings) and outline them in the “Brave Blossom Capital” tab of the valuation spreadsheet.

Workforce

Long term manning requirements for the Brave Blossom Underground are projected to be ~800 staff and contractors. The current workforce utilised at Springbok would be suitably skilled to transition across to Brave Blossom following the completion of mining however due to the short remaining mine life a significant proportion of the workforce is contracted labour. Considerations should be made for if the project is owner operator or contracted out.

Environment & Community

The mine is anticipated to have minimal additional environmental limitations imposed on it relating directly to the underground workings. The Brave Blossom project is not expected to increase the impacts on the community, who will appreciate the extended mine life and support it will provide.

Mine Closure & Rehabilitation

Based on final landform designs, the rehabilitation of the pit void and infrastructure following mining is estimated to cost \$2.5M AUD. This cost does not factor in the legacy rehabilitation and is only an incremental increase on the initial estimate of \$900M AUD. As operations will continue, the closure liability is deferred until later years following completion of Brave Blossom underground.

Marketing

From the limited drilling, the coal quality is expected to have deteriorated compared to Springbok Coal’s Premium Hard Coking Coal with a similar yield to current operations. It is anticipated that long-term customers will provide steady demand for this product at a 12% discount to the current contracted price.

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Quality Parameter	Unit	Basis	Springbok Coal
Moisture	%	ad	2.5
Product Ash	%	ad	8.5
Volatile Matter	%	ad	36
Total Moisture	%	ar	9.0
Total Sulfur	%	ad	0.5
CSN			7.2
Ultimate Analysis			
Nitrogen	%	daf	2.00
Ash Analysis			
Silicon as SiO ₂	%	db	72.0
Sodium as Na ₂ O	%	db	0.30
Ash Fusion Temp (Reducing Atm)			
Deformation	°C		1300
Spherical	°C		1350
Hemispherical	°C		1450
Flow	°C		1500
Trace Elements			
Arsenic (As)	mg/kg	db	1.5
Boron (B)	mg/kg	db	30-60
Cadmium (Cd)	mg/kg	db	0.08
Cobalt (Co)	mg/kg	db	
Copper (Cu)	mg/kg	db	8-10
Fluorine (F)	mg/kg	db	100
Mercury (Hg)	mg/kg	db	0.07
Nickel (Ni)	mg/kg	db	
Lead (Pb)	mg/kg	db	10-15
Selenium (Se)	mg/kg	db	1.1

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Concept Capital Estimate

Brave Blossom requires significant capital to progress. A historic concept capital estimate has been provided below:

Brave Blossom Capital			2027	2028	2029	2030	2031	2032	2033
Capital Items - Units									
Projects / studies	AUD\$	\$ 5,000	\$5,000	\$5,000	\$2,500	\$2,500	\$0	\$0	\$0
Exploration Drilling	AUD\$	\$ 5,000	\$5,000	\$5,000	\$2,500	\$2,500	\$0	\$0	\$0
ROM Bin Upgrade	AUD\$	\$ 15,000	\$0	\$0	\$0	\$0	\$15,000	\$7,500	\$0
Ventilation Shafts	AUD\$	\$ 20,000	\$0	\$0	\$20,000	\$20,000	\$0	\$0	\$0
Mine infrastructure	AUD\$	\$ 5,000	\$0	\$0	\$5,000	\$5,000	\$0	\$0	\$0
Drift	AUD\$	\$ 12,000	\$0	\$0	\$12,000	\$12,000	\$0	\$0	\$0
Drift Conveyor to ROM Stockpile	AUD\$	\$ 17,000	\$0	\$0	\$17,000	\$17,000	\$0	\$0	\$0
ROM Stockpile	AUD\$	\$ 5,000	\$0	\$0	\$0	\$5,000	\$5,000	\$0	\$0
UG Conveyors Relocation & Reuse	AUD\$	\$ 20,000	\$0	\$0	\$0	\$20,000	\$20,000	\$0	\$0
Longwall	AUD\$	\$ 190,000	\$0	\$0	\$0	\$0	\$0	\$190,000	\$0
Mining Equipment	AUD\$	\$ 69,900	\$0	\$0	\$0	\$0	\$69,900	\$0	\$0
Infrastructure + CHPP EPCM	AUD\$	\$ 24,000	\$0	\$0	\$0	\$0	\$0	\$24,000	\$24,000